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ELECTRIC WORK VEHICLE

Abstract

An electric work vehicle includes a harness configured to supply power from a battery to a motor and a connector provided at an end portion of the harness and connected to the motor. The electric work vehicle includes a lock movable between a retaining position at which the lock retains a connection between the connector and the motor and a release position at which the lock allows the connector to be separated from the motor. The electric work vehicle includes a restrictor provided on an movement path of the lock between the retaining position and the release position to prevent the lock from being move from the retaining position to the release position, and the restrictor is removable.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS [0001] This application claims the benefit of priority to Japanese Patent Application No. 2022-210896 filed on Dec. 27, 2022 and is a Continuation application of PCT Application No. PCT/JP2023/031368 filed on Aug. 30, 2023. The entire contents of each application are hereby incorporated herein by reference.

BACKGROUND OF THE INVENTION
1. Field of the Invention
[0002] The present invention relates to electric work vehicles each including a travel device configured to be driven by a motor.
2. Description of the Related Art
[0003] Electric work vehicles include an electric work vehicle in which a battery is supported by a main frame and a motor that drives a travel device is provided below the battery and supported by the main frame as disclosed in JP 2021-957A, for example.
[0004] In JP 2021-957A, the battery is covered by a hood. The motor is below a lower end portion of the hood in a side view and located outside the hood.
[0005] In an electric work vehicle, a lock may be provided for a connector of a harness to which power is supplied from the battery.
[0006] After the connector of the harness is connected to the motor, it is possible to prevent the connector of the harness from coming off the motor by moving the lock to a retaining position. It is possible to take off the connector of the harness from the motor by moving the lock to a release position.
SUMMARY OF THE INVENTION
[0007] If the electric work vehicle disclosed in JP 2021-957A travels on uneven ground, a branch of a plant or the like may come into contact with the vicinity of the motor, and there is a concern that, if the branch of a plant or the like comes into contact with the lock, the lock may be moved from the retaining position to the release position. There is also a concern that the lock may gradually move from the retaining position to the release position due to vibrations during travel.
[0008] Example embodiments of the present invention prevent a lock provided for a connector of a harness connected to a motor of an electric work vehicle from being moved to a release position.
[0009] An electric work vehicle according to an example embodiment of the present invention includes a hood housing a battery, a motor outside the hood, a travel device configured to be driven by the motor, a harness configured to supply power from the battery to the motor, a connector provided at an end portion of the harness and configured to be connected to the motor, a lock movable between a retaining position at which the lock retains a connection between the connector and the motor and a release position at which the lock allows the connector to be separated from the motor, and a restrictor provided on an movement path of the lock between the retaining position and the release position to prevent the lock from being moved from the retaining position to the release position, wherein the restrictor is removable.
[0010] According to an example embodiment of the present invention, a worker can connect the connector of the harness to the motor, move the lock to the retaining position, and then attach the restrictor to a movement path of the lock.
[0011] In this case, even if the lock moves from the retaining position toward the release position, the lock comes into contact with the restrictor and the movement of the lock is blocked, and accordingly, the lock cannot reach the release position.

[0012] According to an example embodiment of the present invention, it is possible to prevent the lock from being moved from the retaining position to the release position due to a branch of a plant or the like coming into contact with the lock during travel, and to prevent the lock from gradually moving from the retaining position to the release position due to vibrations during travel, and accordingly, it is possible to prevent the connector of the harness from coming off the motor.
[0013] According to an example embodiment of the present invention, when the worker removes the connector of the harness from the motor for maintenance or the like, the worker can move the lock from the retaining position to the release position by removing the restrictor. Therefore, the worker can remove the connector of the harness from the motor without difficulty, and the maintenance or the like can be carried out without hindrance.
[0014] It is preferable that an electric work vehicle according to an example embodiment of the present invention includes a connector to attach the restrictor, and the restrictor is removable by removing the connector with a tool.
[0015] According to this example embodiment of the present invention, the restrictor is attached with the use of the connector, and therefore, the restrictor can be firmly attached.
[0016] When the worker removes the connector of the harness from the motor for maintenance or the like, the worker can remove the restrictor by removing the connector with the tool, and the maintenance or the like can be carried out without hindrance.
[0017] In an example embodiment of the present invention, it is preferable that the release position is spaced apart from the connector farther than the retaining position.
[0018] According to this example embodiment of the present invention, the release position of the lock is spaced apart from the connector of the harness farther than the retaining position, and the lock moved to the release position extends outward from the connector of the harness, and therefore, it is easy for the worker to notice that the lock has been moved to the release position.
[0019] Since it is easy for the worker to notice that the lock has been moved to the release position, the worker can be kept from forgetting to attach the restrictor.
[0020] The “release position of the lock being set to a position spaced apart from the connector of the harness farther than the retaining position” means that the retaining position is closer to the connector of the harness.
[0021] Accordingly, the restrictor can also be provided at a position close to the connector of the harness, and therefore, according to this example embodiment of the present invention, it is possible to save the space occupied by the connector of the harness, the lock moved to the retaining position, and the restrictor.
[0022] Since the lock moved to the retaining position is close to the connector of the harness, the lock moved to the retaining position does not extend outward from the connector of the harness and a branch of a plant or the like is less likely to come into contact with the lock, and this is advantageous in preventing the lock from being moved from the retaining position to the release position.
[0023] In an example embodiment of the present invention, it is preferable that the restrictor extends across an entire width of the lock in the retaining position.
[0024] According to this example embodiment of the present invention, the restrictor extends across the entire width of the lock in the retaining position, and therefore, when the lock moves from the retaining position toward the release position, the restrictor comes into contact with a large area of the lock, and the movement of the lock toward the release position can be stably blocked by the restrictor. This is advantageous in preventing the lock from being moved from the retaining position to the release position.
[0025] In an example embodiment of the present invention, it is preferable that the restrictor includes a first portion extending along a horizontal direction and a pair of second portions extending in a same direction that is an upward direction or a downward direction from an end portion and another end portion of the first portion, the first portion is located above or below the

connector, and the second portions are located laterally outward of the connector.
[0026] According to this example embodiment of the present invention, the restrictor surrounds the connector of the harness and the lock, and the movement of the lock toward the release position can be stably blocked by the restrictor, and this is advantageous in preventing the lock from being moved from the retaining position to the release position.
[0027] It is preferable that the electric work vehicle according to an example embodiment of the present invention includes a cover under the harness and the connector to cover the harness and the connector, the lock is located above the connector, the first portion is located above the connector, and lower portions of the second portions are attached to the cover.
[0028] According to this example embodiment of the present invention, the harness and the connector are covered by the cover from below, and therefore, flying gravel and the like can be kept from hitting the harness and the connector.
[0029] According to this example embodiment of the present invention, the lock is located above the connector of the harness. In the restrictor including the first portion and the second portions as described above, the first portion is located above the connector of the harness, and the lower portions of the second portions are attached to the cover.
[0030] The cover covering the harness and the connector from below is effectively used to attach the restrictor, and the cover has both a function of protecting the harness and the connector and a function of supporting the restrictor, and therefore, the structure can be simplified due to the cover having the plurality of functions.
[0031] In an example embodiment of the present invention, it is preferable that the cover includes an opening to enable cleaning or inspection, the opening is closed with a covering attached to the cover, and the covering is removable from the cover.
[0032] According to this example embodiment of the present invention, it is possible to visually inspect the harness and the connector through the opening in the cover by removing the covering. Even if soil or the like accumulates on the cover, it is possible to take off the soil or the like that has accumulated on the cover from the opening in the cover by removing the covering.
[0033] It is preferable that the electric work vehicle according to an example embodiment of the present invention includes left and right main frames extending along a front-rear direction, the connector and the lock are between the left and right main frames and overlap the left and right main frames in a side view, and the restrictor is attached to extend between the left and right main frames.
[0034] According to this example embodiment of the present invention, when the harness and the connector are between the right and left main frames, the restrictor is attached to extend between the right and left main frames, and accordingly, the main frames have a function of supporting the restrictor, and the structure can be simplified due to the main frames having a plurality of functions.
[0035] The above and other elements, features, steps, characteristics and advantages of the present invention will become more apparent from the following detailed description of the example embodiments with reference to the attached drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0036] FIG. **1** is a left side view of a tractor.
[0037] FIG. **2** is a plan view of the tractor.
[0038] FIG. **3** is a schematic diagram showing a transmission system for transmission from a motor to front wheels and rear wheels.
[0039] FIG. **4** is a left side view showing the inside of a hood.
[0040] FIG. **5** is a left side view showing the vicinity of an inverter and the motor.

[0041] FIG. **6** is a front view of a vertical cross section showing the vicinity of a support frame and the inverter.
[0042] FIG. **7** is a front view of a vertical cross section showing the vicinity of the motor.
[0043] FIG. **8** is an exploded perspective view of the support frame and the motor.
[0044] FIG. **9** is a plan view of the support frame.
[0045] FIG. **10** is a left side view of the support frame.
[0046] FIG. **11** is a front view of the support frame.
[0047] FIG. **12** is a plan view showing the vicinity of a terminal of the motor, a connector, and a restrictor.
[0048] FIG. **13** is a left side view of a vertical cross section showing the vicinity of the terminal of the motor, the connector, and the restrictor.
[0049] FIG. **14** is a left side view of a vertical cross section showing the vicinity of the terminal of the motor, the connector, and the restrictor in a variation of an example embodiment of the present invention.

DETAILED DESCRIPTION OF THE EXAMPLE EMBODIMENTS

[0050] FIGS. **1** to **14** show tractors that are examples of electric work vehicles. In FIGS. **1** to **14**, F indicates the forward direction, B indicates the backward direction, U indicates the upward direction, D indicates the downward direction, R indicates the rightward direction, and L indicates the leftward direction.
[0051] As shown in FIGS. **1** and **2**, a body **3** is supported by right and left front wheels **1** (corresponding to a travel device) and right and left rear wheels **2** (corresponding to a travel device). A hood **4** is provided in a front portion of the body **3**, and a driving section **5** is provided in a rear portion of the body **3**. A steering wheel **6** for steering the front wheels **1**, a driver's seat **7**, a floor section **8**, and a ROPS frame **9** are provided in the driving section **5**.
[0052] The body **3** includes right and left main frames **10**, a continuously variable transmission case **11**, a transmission case **12**, and the like. The continuously variable transmission case **11** is coupled to a front portion of the transmission case **12**, and the right and left main frames **10** are coupled to the continuously variable transmission case **11** and the transmission case **12** and extend along the front-rear direction. The front wheels **1** are supported by the main frames **10**, and the rear wheels **2** are supported by the transmission case **12**.
[0053] With this configuration, the tractor includes the front wheels **1**, the rear wheels **2**, and the main frames **10** supporting the front wheels **1**.
[0054] As shown in FIG. **1**, a battery **13**, an inverter **14**, and a motor **15** are supported by the main frames **10**.
[0055] As shown in FIG. **4**, a connecting portion **16** is provided in a front left portion of the battery **13**, and a charger (not shown) of an external power source (not shown) is connected to the connecting portion **16** to charge the battery **13** with power supplied from the external power source via the connecting portion **16**. Power is supplied from the battery **13** to the inverter **14**, and the DC power supplied from the battery **13** is converted to AC power in the inverter **14** and supplied to the motor **15** to drive the motor **15**.
[0056] As shown in FIGS. **2** and **4**, a DC-DC converter **17**, a battery **18**, a radiator **19**, a water pump **20**, and the like are provided on front portions of the main frames **10**. The battery **13**, the inverter **14**, the connecting portion **16**, the DC-DC converter **17**, the battery **18**, the radiator **19**, and the like are covered by the hood **4**.
[0057] With this configuration, the tractor includes the battery **13** and the hood housing the battery **13**.
[0058] Power is supplied from the battery **13** to the DC-DC converter **17** and the voltage is reduced to 12 V, for example, in the DC-DC converter **17**. The power having a voltage reduced to 12 V is supplied to the battery **13**, and then supplied from the battery **13** to the battery **18**. Various devices (not shown) included in the tractor move with the power with the voltage of 12 V supplied from the

battery **18**.

[0059] When the water pump **20** moves, cooling water generated in the radiator **19** is supplied to the water pump **20** via a pipe **21**, and supplied from the water pump **20** to the inverter **14**. The cooling water that has passed through the inverter **14** is supplied to the motor **15** via a pipe **22**, and the cooling water that has passed through the motor **15** is supplied to the DC-DC converter **17**, and returns from the DC-DC converter **17** to the radiator **19** via a pipe **23**.

[0060] As shown in FIGS. **1** and **3**, a hydraulic continuously variable transmission **24** is housed in the continuously variable transmission case **11**, and motive power is transmitted from the motor **15** to the continuously variable transmission **24** via a transmission shaft **25**. The continuously variable transmission **24** is capable of changing the speed of forward and rearward movements in a stepless manner, and is moved via a gearshift pedal **26** (see FIG. **2**) provided on the floor section **8**.

[0061] A sub transmission **27**, a rear-wheel differential device **28**, a front-wheel transmission **29**, and right and left brakes **30** are housed in the transmission case **12**. Motive power is transmitted from the continuously variable transmission **24** to the sub transmission **27**, and transmitted from the sub transmission **27** to the rear wheels **2** via the rear-wheel differential device **28**.

[0062] Motive power diverging between the sub transmission **27** and the rear-wheel differential device **28** is transmitted to the front-wheel transmission **29**, transmitted from the front-wheel transmission **29** to a front-wheel differential device **32** via a transmission shaft **31**, and then transmitted from the front-wheel differential device **32** to the front wheels **1**.

[0063] When the front wheels **1** are moved to be within the ranges of right and left set angles from a straight travel position, the front wheels **1** are driven by the front-wheel transmission **29** at the same speed as the speed of the rear wheels **2**. When the front wheels **1** are steered rightward or leftward past the right or left set angle, the front wheels **1** are driven by the front-wheel transmission **29** at a speed higher than the speed of the rear wheels **2**.

[0064] With this configuration, the tractor includes the motor **15** that moves with power supplied from the battery **13** and drives the front wheels **1** and the rear wheels **2**, and the front wheels **1** and the rear wheels **2** driven by the motor **15**.

[0065] The right brake **30** is provided on a transmission system for transmission from the rear-wheel differential device **28** to the right rear wheel **2**, and the left brake **30** is provided on a transmission system for transmission from the rear-wheel differential device **28** to the left rear wheel **2**. A right brake pedal **33** that can move the right brake **30** and a left brake pedal **33** that can move the left brake **30** are provided in a left front portion **8***b* of the floor section **8** (see FIG. **2**).

[0066] It is possible to make a turn with a small turning radius by stepping on the right (left) brake pedal **33** corresponding to the center of turning when making a right turn (left turn) to apply the right (left) brake **30**.

[0067] As shown in FIGS. **4** to **7**, a support frame **34** is attached to the main frames **10**, and the battery **13** is placed on and coupled to the support frame **34**.

[0068] As shown in FIGS. **8** to **11**, the support frame **34** includes front-rear frames **35** and **36** (corresponding to base portions), left-right frames **37**, **38**, and **39** (corresponding to base portions), right and left leg portions **40**, right and left coupling plates **41**, right and left brackets **42**, upright portions **45** and **46**, and the like.

[0069] The front-rear frames **35** and **36** are angled bars extending along the front-rear direction. The left-right frame **37** is a flat plate extending along the left-right direction. The left-right frames **38** and **39** are angled bars extending along the left-right direction. The front-rear frame **35** and the right front-rear frame **36** are spaced apart from each other in the left-right direction. Also, the front-rear frame **35** and the left front-rear frame **36** are spaced apart from each other in the left-right direction. Accordingly, the front-rear frame **35**, the right front-rear frame **36**, and the left front-rear frame **36** are spaced apart from each other in the left-right direction. The left-right frame **37**, the left-right frame **38**, and the left-right frame **39** are spaced apart from each other in the front-rear direction. The front-rear frames **35** and **36** are coupled to the left-right frames **37**, **38**, and **39**.

Accordingly, the support frame **34** (the front-rear frames **35** and **36** and the left-right frames **37**, **38**, and **39**) has a lattice shape in a plan view.
[0070] Two channel-shaped leg portions **40** are coupled to a front portion of the right front-rear frame **36**. A coupling plate **41** extending in the front-rear direction is coupled to lower portions of the two leg portions **40**. Two channel-shaped leg portions **40** are coupled to a front portion of the left front-rear frame **36**. A coupling plate **41** extending in the front-rear direction is coupled to lower portions of the two leg portions **40**. The right bracket **42** is coupled to a rear portion of the right front-rear frame **36**, and the left bracket **42** is coupled to a rear portion of the left front-rear frame **36**.
[0071] The upright portion **45** having a flat plate shape is coupled to the left-right frame **37** with bolts and extends upward from the left-right frame **37**. The right upright portion **46** having a flat plate shape is coupled to the right front-rear frame **36** with bolts. Also, the left upright portion **46** having a flat plate shape is coupled to the left front-rear frame **36** with bolts. Accordingly, the left and right upright portions **46** extend upward from the left and right front-rear frames **36**.
[0072] As shown in FIGS. **4** to **7**, a right support **43** having a flat plate shape is coupled to a front upper portion of the right main frame **10** along the front-rear direction. Also, a left support **43** having a flat plate shape is coupled to a front upper portion of the left main frame **10** along the front-rear direction. A gear case **44** is coupled to the right and left main frames **10** along the left-right direction.
[0073] The right leg portions **40** and the right coupling plate **41** of the support frame **34** are coupled to the right support **43** with bolts, and the left leg portions **40** and the left coupling plate **41** of the support frame **34** are coupled to the left support **43** with bolts. The right bracket **42** of the support frame **34** is coupled to a right portion of the gear case **44** with bolts, and the left bracket **42** of the support frame **34** is coupled to a left portion of the gear case **44** with bolts.
[0074] As described above, the support frame **34** is attached to the main frames **10**, and the front-rear frames **36** of the support frame **34** are located above upper ends of the main frames **10** and slightly spaced apart therefrom.
[0075] As shown in FIGS. **4** to **7**, the battery **13** is placed on the front-rear frames **35** and **36** and the left-right frames **37**, **38**, and **39** of the support frame **34**, and an outer periphery of a lower portion of the battery **13** is surrounded by the upright portions **45** and **46** of the support frame **34**.
[0076] A lower front portion of the battery **13** is coupled to the upright portion **45** with bolts, a lower right portion and a lower left portion of the battery **13** are coupled to the upright portions **46** with bolts, and a lower rear portion of the battery **13** is coupled to the left-right frame **39** with bolts to couple the battery **13** to the support frame **34**.
[0077] With this configuration, the tractor includes the support frame **34** that is attached to the main frames **10** and on which the battery **13** is placed to be supported by the main frames **10**.
[0078] The support frame **34** includes the plurality of front-rear frames **35** and **36** extending along the front-rear direction and spaced apart from each other in the left-right direction and the plurality of left-right frames **37**, **38**, and **39** extending along the left-right direction and spaced apart from each other in the front-rear direction.
[0079] The support frame **34** includes the front-rear frames **35** and **36** and the left-right frames **37**, **38**, and **39** on which the battery is placed and the upright portions **45** and **46** extending upward from the front-rear frames **35** and **36** and the left-right frames **37**, **38**, and **39** and surrounding the outer periphery of the lower portion of the battery **13**.
[0080] As shown in FIGS. **8**, **9**, and **10**, a coupling portion **47** is provided in a rear portion (between the left-right frame **38** and the left-right frame **39**) of the front-rear frame **35** of the support frame **34**.
[0081] The coupling portion **47** is formed by bending a plate into a channel shape, and is provided on the support frame **34** by being coupled to a lower surface of the front-rear frame **35** of the support frame **34** to protrude downward from the front-rear frame **35** of the support frame **34**.

[0082] A coupling bolt **48** is inserted into an opening in the coupling portion **47**, and a portion of the front-rear frame **35** of the support frame **34** above the coupling portion **47** includes an opening **35***a*. A worker can fasten the coupling bolt **48** and remove the coupling bolt **48** with a tool such as a wrench by inserting the tool into the opening **35***a* of the front-rear frame **35**.
[0083] As shown in FIGS. **4**, **5**, and **7** to **10**, the motor **15** is located under the rear portion (between the left-right frames **38** and **39**) of the front-rear frame **35** of the support frame **34** and between the right and left main frames **10**. The motor **15** is coupled to the coupling portion **47** with the coupling bolt **48**, and a rear portion of the motor **15** is coupled to the gear case **44** with bolts.
[0084] With this configuration, the coupling portion **47** that protrudes downward and to which the motor **15** is coupled is provided on the support frame **34**, and the motor **15** is supported by the support frame **34** by being suspended from the support frame **34**. A portion of the support frame **34** above the coupling portion **47** includes the opening **35***a* that exposes the coupling portion **47** upward, and the opening **35***a* is provided in the front-rear frame **35** of the support frame **34**.
[0085] An upper portion of the motor **15** is coupled to the support frame **34** (coupling portion **47**). The rear portion of the motor **15** is coupled to the main frames **10** via the gear case **44**, and the rear portion of the motor **15** is coupled to the continuously variable transmission case **11** and the transmission case **12** via the gear case **44**.
[0086] The motor **15** is located below a lower end portion of the hood **4** and the support frame **34** (front-rear frames **36**) in a side view (see FIGS. **1** and **4**), and the motor **15** can be seen through a gap between upper end portions of the main frames **10** and the lower end portion of the hood **4** and the support frame **34** (front-rear frames **36**).
[0087] With this configuration, the tractor includes the motor **15** located outside the hood **4**.
[0088] As shown in FIGS. **4**, **5**, and **6**, the inverter **14** is supported by the main frames **10** by being coupled to the right and left supports **43** with bolts, and is located at a position next to the motor **15** in the front-rear direction and under a front portion (between the left-right frames **37** and **38**) of the support frame **34**.
[0089] As shown in FIGS. **8** and **11**, a lower portion **45***a* of the upright portion **45** (corresponding to a portion of the support frame **34** adjacent to the inverter **14**) is recessed upward, and the upright portion **45** has a gate shape in a front view.
[0090] When the DC-DC converter **17**, the battery **18**, the radiator **19**, and the like are removed, the worker can see the inverter **14** from the front side through a region under the lower portion **45***a* of the upright portion **45** as shown in FIG. **6**.
[0091] In the state where the DC-DC converter **17**, the battery **18**, the radiator **19**, and the like are removed, the worker can remove the inverter **14** from the main frames **10** by removing the bolts coupling the inverter **14** to the supports **43** and moving the inverter **14** forward along the horizontal direction through the region under the lower portion **45***a* of the upright portion **45**.
[0092] After maintenance of the inverter **14** or the like is finished, the worker can place the inverter **14** on the supports **43** by moving the inverter **14** rearward along the horizontal direction from a position in front of the main frames **10** through the region under the lower portion **45***a* of the upright portion **45** and couple the inverter **14** to the supports **43** with bolts.
[0093] With this configuration, the inverter **14** that supplies power from the battery **13** to the motor **15** is provided next to the motor **15** under the support frame **34**.
[0094] The upright portion **45** of the support frame **34** adjacent to the inverter **14** is recessed upward to allow movement of the inverter **14** in the horizontal direction.
[0095] The upright portion **45** of the support frame **34** adjacent to the inverter **14** is a portion of the support frame **34** located on a side opposite to the motor **15** with respect to the inverter **14**.
[0096] As shown in FIGS. **1** and **2**, the driving section **5** and the floor section **8** of the driving section **5** are provided behind the battery **13** and the support frame **34**. A right front portion **8***a* and the left front portion **8***b* of the floor section **8** extend forward and diagonally upward to positions adjacent to right rear portions and left rear portions of the battery **13** and the support frame **34**.

[0097] As shown in FIGS. **8** to **11**, right and left floor frames **49** are provided. Each floor frame **49** includes a base plate portion **49***a* having a flat plate shape and a support portion **49***b* having a channel shape and coupled to the base plate portion.
[0098] The base plate portions **49***a* of the right and left floor frames **49** are coupled to a right rear portion and a left rear portion of the support frame **34** (front-rear frames **36**) with bolts **50**. As shown in FIG. **7**, the right front portion **8***a* and the left front portion **8***b* of the floor section **8** are supported by the support portions **49***b* of the right and left floor frames **49** via anti-vibration members **51**. The floor frames **49** can be removed from the support frame **34** (front-rear frames **36**) by removing the bolts **50**.
[0099] With this configuration, the tractor includes the right and left floor frames **49** capable of supporting the right front portion **8***a* and the left front portion **8***b* of the floor section **8**, and the right and left floor frames **49** are attachable to and detachable from the right rear portion and the left rear portion of the support frame **34**.
[0100] As shown in FIGS. **4** and **5**, the tractor includes a harness **52**, and a connector **53** is provided at a front end portion of the harness **52**, and the connector **53** is connected to a terminal **14***a* provided in a front lower portion of the inverter **14**. A connector **54** is provided at a rear end portion of the harness **52**, and the connector **54** is connected to a terminal **15***a* provided in a front portion of the motor **15**.
[0101] As shown in FIGS. **5**, **6**, and **7**, the harness **52**, the connectors **53** and **54**, the terminal **14***a* of the inverter **14**, and the terminal **15***a* of the motor **15** are positioned between the right and left main frames **10** to overlap the right and left main frames **10** in a side view. A cover **55** having a flat plate shape is coupled to lower portions of the right and left main frames **10** to extend between front end portions of the main frames **10** and a position under the terminal **15***a* of the motor **15**.
[0102] As shown in FIGS. **1** and **4**, a front axle case **56** supporting the front wheels **1** is supported by a front portion of the cover **55**, and the transmission shaft **31** (see FIG. **3**) extends forward under the cover **55** and is connected to the front-wheel differential device **32** (see FIG. **3**) housed in the front axle case **56**. The water pump **20** described above is attached to an upper surface of the front portion of the cover **55**.
[0103] As shown in FIGS. **5** and **7**, an opening **55***a* to enable cleaning or inspection is provided in a portion of the cover **55** between the motor **15** and the front axle case **56** (between the connectors **53** and **54**). A covering **57** having a flat plate shape is provided, and the opening **55***a* of the cover **55** is closed with the covering **57** by attaching the covering **57** to the cover **55** with bolts **58**. It is possible to remove the covering **57** from the cover **55** by removing the bolts **58** to open the opening **55***a* of the cover **55**.
[0104] With this configuration, DC power supplied from the battery **13** is converted to AC power in the inverter **14**, and then supplied from the inverter **14** to the motor **15** via the harness **52**. Thus, the tractor includes the harness **52** to supply power from the battery **13** to the motor **15**. The tractor includes the connector **54** provided at an end portion of the harness **52** and connected to the motor **15**.
[0105] The cover **55** covering the harness **52**, the connectors **53** and **54**, the terminal **14***a* of the inverter **14**, and the terminal **15***a* of the motor **15** is provided under the harness **52** and the connectors **53** and **54**.
[0106] The cover **55** includes the opening **55***a* to enable cleaning or inspection. The covering **57** is attached to the cover **55** to close the opening **55***a*, and the covering **57** is removable from the cover **55**.
[0107] As shown in FIGS. **12** and **13**, a lock **59** is provided on the connector **54** on the upper side of the connector **54**. As shown in FIGS. **5** and **7**, the worker can see the connector **54** and the lock **59** through the gap between the lower end portion of the hood **4** and the upper end portions of the main frames **10** by looking diagonally downward from positions above right and left sides of the body **3**.

[0108] The lock **59** is supported by the connector **54** to be pivotable about an axis P**1** extending along the left-right direction and is movable between a retaining position A**1** at which the lock retains connection between the connector **54** and the terminal **15***a* of the motor **15** and a release position A**2** at which the lock allows the connector **54** to be separated from the terminal **15***a* of the motor **15**.
[0109] The retaining position A**1** is at which the lock is substantially in contact with an upper portion of the connector **54** and that is spaced apart from the motor **15** further forward than the release position A**2** is. The release position A**2** is spaced apart from the connector **54** further upward than the retaining position A**1** is and is closer to the motor **15** than the retaining position A**1** is.
[0110] When the worker connects the connector **54** to the terminal **15***a* of the motor **15**, the worker connects the connector **54** to the terminal **15***a* of the motor **15** with the lock **59** moved to the release position A**2**, and then moves the lock **59** to the retaining position A**1**.
[0111] When the worker removes the connector **54** from the terminal **15***a* of the motor **15**, the worker moves the lock **59** from the retaining position A**1** to the release position A**2**, and then moves the connector **54** forward from the terminal **15***a* of the motor **15** to remove the connector.
[0112] As shown in FIGS. **4** and **5**, a lock **59** is provided on the connector **53** in the same manner as the connector **54**, on the lower side of the connector **53**. The state shown in FIG. **5** is a state where the lock **59** on the connector **53** is moved to the retaining position A**1**, and the lock **59** can be moved to the release position A**2** by moving the lock **59** downward and forward.
[0113] With this configuration, the tractor includes the lock **59** movable between the retaining position A**1** at which the lock retains connection between the connector **54** and the motor **15** and the release position A**2** at which the lock allows the connector **54** to be separated from the motor **15**.
[0114] The lock **59** is provided on the upper side of the connector **54**, and the release position A**2** is spaced apart from the connector **54** farther than the retaining position A**1**.
[0115] As shown in FIGS. **7**, **12**, and **13**, a restrictor **60** is provided for the connector **54**. The restrictor **60** includes a first portion **60***a* extending along a horizontal direction and a pair of second portions **60***b* extending in the same direction that is the upward direction or the downward direction from an end portion and another end portion of the first portion **60***a*.
[0116] The state shown in FIGS. **7**, **12**, and **13** is a state where lower portions of the second portions **60***b* of the restrictor **60** are attached to the cover **55** with bolts **61** (corresponding to connectors) and the restrictor **60** is attached to a restricting position B**1**.
[0117] In the state where the restrictor **60** is attached to the restricting position B**1**, the cover **55** covering the restrictor **60** is under the restrictor **60**, and the restrictor **60** is located between the right and left main frames **10**, overlapping the right and left main frames **10** in a side view.
[0118] In the state where the restrictor **60** is attached to the restricting position B**1**, the first portion **60***a* of the restrictor **60** extends along the left-right direction above the connector **54** and the lock **59** moved to the retaining position A**1** across the entire widths of the connector **54** and the lock **59** moved to the retaining position A**1**. The second portions **60***b* of the restrictor **60** extend along the up-down direction at right and left positions laterally outward of the connector **54** and the lock **59** moved to the retaining position A**1**.
[0119] In the state where the restrictor **60** is attached to the restricting position B**1**, the first portion **60***a* of the restrictor **60** is on an movement path of the lock **59** between the retaining position A**1** and the release position A**2**. Therefore, even if the lock **59** moves from the retaining position A**1** toward the release position A**2**, the lock **59** comes into contact with the restrictor **60** (the first portion **60***a*) and the movement of the lock **59** is blocked, and accordingly, the lock **59** cannot reach the release position A**2**.
[0120] The worker can remove the restrictor **60** from the cover **55** (the restricting position B**1**) by removing the bolts **61** with a tool (not shown) such as a wrench. When the restrictor **60** is removed

from the cover **55** (the restricting position B**1**), the worker can move the lock **59** from the retaining position A**1** to the release position A**2** and remove the connector **54** from the terminal **15***a* of the motor **15**.
[0121] With this configuration, the tractor includes the restrictor **60** provided on the movement path of the lock **59** between the retaining position A**1** and the release position A**2** to hinder the lock **59** from being moved from the retaining position A**1** to the release position A**2**, and the restrictor **60** is removable. The restrictor **60** extends across the entire width of the lock **59** moved to the retaining position A**1**. The bolts **61** (connectors) to attach the restrictor **60** are provided, and the restrictor **60** can be removed by removing the bolts **61** (connectors) with a tool.
[0122] The restrictor **60** includes the first portion **60***a* extending along the horizontal direction and the pair of second portions **60***b* extending in the same direction that is the upward direction or the downward direction from an end portion and another end portion of the first portion **60***a*. The first portion **60***a* is located above the connector **54**, the second portions **60***b* are located laterally outward of the connector **54**, and the lower portions of the second portions **60***b* are attached to the cover **55**.
First Variation of an Example Embodiment of the Present Invention
[0123] A configuration is also possible in which the coupling portion **47** is provided on one of the front-rear frames **36** of the support frame **34** and the opening **35***a* is provided in a portion of the front-rear frame **36** of the support frame **34** above the coupling portion **47**. A configuration is also possible in which a plurality of coupling portions **47** are provided on the front-rear frames **35** and **36** and the left-right frames **37**, **38**, and **39** of the support frame **34** and openings **35***a* are provided in portions of the front-rear frames **35** and **36** and the left-right frames **37**, **38**, and **39** above the coupling portions **47**.
[0124] With this configuration, the opening **35***a* is provided in any of the plurality of front-rear frames **35** and **36** and the plurality of left-right frames **37**, **38**, and **39**.
Second Variation of an Example Embodiment of the Present Invention
[0125] The coupling portion **47** may be omitted.
[0126] In this configuration, the motor **15** can be directly coupled to the support frame **34** (the front-rear frames **35** and **36** and the left-right frames **37**, **38**, and **39**) with the coupling bolt **48**. The motor **15** can be supported by the support frame **34** (the front-rear frames **35** and **36** and the left-right frames **37**, **38**, and **39**) by being suspended with use of a suspension member (not shown).
Third Variation of an Example Embodiment of the Present Invention
[0127] A configuration is also possible in which the gear case **44** coupled to the right and left main frames **10** is in front of the motor **15**, the upper portion of the motor **15** is coupled to the coupling portion **47**, and a front portion of the motor **15** is coupled to the main frames **10** via the gear case **44**.
[0128] A configuration is also possible in which the positional relationship between the inverter **14** and the motor **15** is reversed, the motor **15** is under the front portion of the support frame **34**, and the inverter **14** is under the rear portion of the support frame **34**.
Fourth Variation of an Example Embodiment of the Present Invention
[0129] The upright portions **45** and **46** of the support frame **34** may be omitted.
[0130] In this configuration, the battery **13** can be placed on the support frame **34** and coupled to the front-rear frames **35** and **36** and the left-right frames **37**, **38**, and **39** of the support frame **34** with bolts.
Fifth Variation of an Example Embodiment of the Present Invention
[0131] A configuration is also possible in which lower portions of the upright portions **46** and lower portions of the front-rear frames **36** of the support frame **34** are recessed upward.
[0132] In this configuration, the upright portions **46** and the front-rear frames **36** of the support frame **34** are portions of the support frame **34** adjacent to the inverter **14**.
[0133] It is possible to remove the inverter **14** from the main frames **10** by removing the right or left leg portions **40** and the coupling plate **41** of the support frame **34** without removing the DC-DC

converter **17**, the battery **18**, the radiator **19**, and the like, and by moving the inverter **14** rightward or leftward along the horizontal direction through a region under the lower portion of the upright portion **46** and the lower portion of the front-rear frame **36**.

Sixth Variation of an Example Embodiment of the Present Invention

[0134] As shown in FIG. **14**, the restrictor **60** may also have a straight bar shape.

[0135] In this configuration, the restrictor **60** is attached to the restricting position B**1** by attaching a right end portion and a left end portion of the restrictor **60** to the right and left main frames **10** with the bolts **61**.

[0136] With this configuration, the harness **52**, the connectors **53** and **54**, the terminal **14***a* of the inverter **14**, and the terminal **15***a* of the motor **15** are between the right and left main frames **10** to overlap the right and left main frames **10** in a side view, and the restrictor **60** is attached to extend between the left and right main frames **10**.

Seventh Variation of an Example Embodiment of the Present Invention

[0137] The restrictor **60** shown in FIGS. **7**, **12**, and **13** and the restrictor **60** of the sixth variation of an example embodiment of the present invention may also be configured as described below, rather than being completely removed from the cover **55** (the main frames **10**).

[0138] The restrictor **60** is configured such that the lock **59** can be moved from the retaining position A**1** to the release position A**2** when an end portion (one second portion **60***b*) of the restrictor **60** is removed and the position of the restrictor **60** is changed with the other end portion (the other second portion **60***b*) of the restrictor **60** kept attached to the cover **55** (the main frames **10**). In other words, the restrictor **60** can be removed from the movement path of the lock **59**.

[0139] In this configuration, the restrictor **60** is removed from the restricting position B**1** by changing the position of the restrictor **60** from the restricting position B**1**. Since the other end portion (the other second portion **60***b*) of the restrictor **60** is kept attached to the cover **55** (the main frames **10**), the restrictor **60** will not be lost.

Eighth Variation of an Example Embodiment of the Present Invention

[0140] A configuration is also possible in which the positional relationship between the retaining position A**1** and the release position A**2** of the lock **59** shown in FIGS. **12** and **13** is reversed, and the release position A**2** is at which the lock is substantially in contact with the upper portion of the connector **54**, and the retaining position A**1** is spaced apart from the connector **54** further upward than the release position A**2** is.

[0141] In this configuration, the restrictor **60** can be configured to be located between the connector **54** and the lock **59** moved to the retaining position A**1** when the restrictor **60** is attached to the restricting position B**1**.

[0142] In this case, even if the lock **59** moves from the retaining position A**1** toward the release position A**2** to be closer to the connector **54**, the lock **59** comes into contact with the restrictor **60** and the movement of the lock **59** is blocked, and accordingly, the lock **59** cannot reach the release position A**2**.

Ninth Variation of an Example Embodiment of the Present Invention

[0143] The lock **59** shown in FIGS. **12** and **13** may be provided on the lower side of the connector **54**.

[0144] The locks **59** of the sixth through eighth variations of example embodiments of the present invention may be provided on the lower side of the connector **54**.

[0145] In this configuration, the restrictor **60** can be provided on the lower side of the connector **54** and the lock **59** as shown in the sixth variation of an example embodiment of the present invention and FIG. **14**.

Tenth Variation of an Example Embodiment of the Present Invention

[0146] The lock **59** shown in FIGS. **12** and **13** may also be provided on the motor **15**.

[0147] The locks **59** of the sixth through ninth variations of example embodiments of the present invention may also be provided on the motor **15**.

[0148] In this configuration, when the connector **54** is removed from the terminal **15***a* of the motor **15**, the lock **59** remains on the motor **15**.

Eleventh Variation of an Example Embodiment of the Present Invention

[0149] The lock **59** shown in FIGS. **12** and **13** may also be configured to be slid along a horizontal direction or the up-down direction between the retaining position A**1** and the release position A**2**. The locks **59** of the sixth through tenth variations of example embodiments of the present invention may also be configured to be slid along a horizontal direction or the up-down direction between the retaining position A**1** and the release position A**2**.

Twelfth Variation of an Example Embodiment of the Present Invention

[0150] The configuration of the lock **59** shown in FIGS. **12** and **13** may also be applied to the lock **59** of the connector **53** (see FIG. **5**) connected to the terminal **14***a* of the inverter **14**. The configurations of the locks **59** of the sixth through eleventh variations of example embodiments of the present invention may also be applied to the lock **59** of the connector **53** (see FIG. **5**) connected to the terminal **14***a* of the inverter **14**.

Thirteenth Variation of an Example Embodiment of the Present Invention

[0151] The tractor may include a crawler device (not shown) as a travel device instead of the rear wheels **2**.

[0152] The tractor may include crawler devices (not shown) as travel devices instead of the front wheels **1** and the rear wheels **2**.

[0153] Example embodiments of the present invention are applicable not only to tractors but also to other electric work vehicles such as electric carts, transport vehicles, and the like.

[0154] While example embodiments of the present invention have been described above, it is to be understood that variations and modifications will be apparent to those skilled in the art without departing from the scope and spirit of the present invention. The scope of the present invention, therefore, is to be determined solely by the following claims.

Claims

1. An electric work vehicle comprising: a hood housing a battery; a motor outside the hood; a travel device configured to be driven by the motor; a harness configured to supply power from the battery to the motor; a connector provided at an end portion of the harness and configured to be connected to the motor; a lock movable between a retaining position at which the lock retains a connection between the connector and the motor and a release position at which the lock allows the connector to be separated from the motor; and a restrictor provided on an movement path of the lock between the retaining position and the release position to prevent the lock from being moved from the retaining position to the release position; wherein the restrictor is removable.

2. The electric work vehicle according to claim 1, further comprising: a connector to attach the restrictor; wherein the restrictor is removable by removing the connector with a tool.

3. The electric work vehicle according to claim 1, wherein the release position is spaced apart from the connector farther than the retaining position.

4. The electric work vehicle according to claim 1, wherein the restrictor extends across an entire width of the lock in the retaining position.

5. The electric work vehicle according to claim 1, wherein the restrictor includes a first portion extending along a horizontal direction and a pair of second portions extending in a same direction that is an upward direction or a downward direction from an end portion and another end portion of the first portion; and the first portion is located above or below the connector, and the second portions are located laterally outward of the connector.

6. The electric work vehicle according to claim 5, further comprising: a cover provided under the harness and the connector to cover the harness and the connector; wherein the lock is located above the connector; and the first portion is located above the connector, and lower portions of the second portions are attached to the cover.

7. The electric work vehicle according to claim 6, wherein the cover includes an opening to enable cleaning or inspection; the opening is closed with a covering attached to the cover; and the covering is removable from the cover.

8. The electric work vehicle according to claim 1, further comprising: left and right main frames extending along a front-rear direction; wherein the connector and the lock are between the left and right main frames and overlap the left and right main frames in a side view; and the restrictor is attached to extend between the left and right main frames.